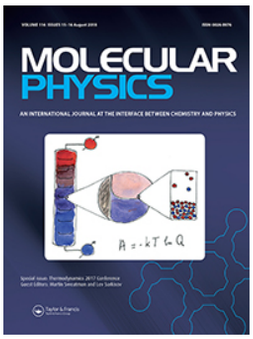

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Multiscale modelling strategies and experimental insights for the solvation of cellulose and hemicellulose in ionic liquids

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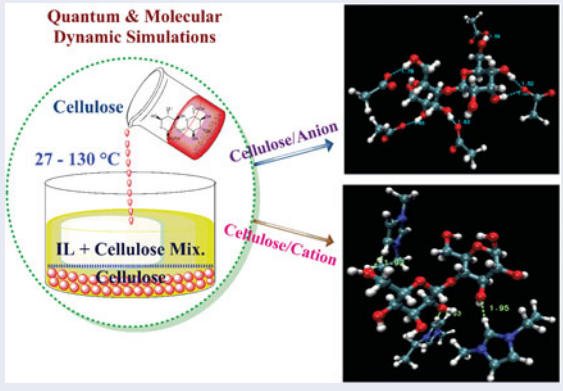
ABSTRACT

The present study investigates the dissolution behaviour of cellulose and hemicellulose in potential ionic liquids (ILs) using both the quantum chemical and experimental validation. For converging upon the recommended IL, 1428 ILs consisting of 42 cations and 34 anions were studied with the conductor like screening model for real solvents (COSMO-RS) model. Based on the infinite dilution activity coefficient of the components in IL, the selected anions and cations were visualised by observing their interactions with cellulose and hemicellulose using interaction energies, natural bonding orbital analysis and molecular dynamics simulations. The dissolution order of cellulose and hemicellulose in ILs was primarily determined by the evaluation of hydrogen bonds between the oxygen atom of anion and hydroxyl proton of cellulose/hemicellulose. From this discernible fact, the anion of the IL was observed to play a leading role in the solvation process as compared to the cation. Eventually, acetate $[OAc]^-$ anion and 1-ethyl-3-methylimidazolium $[EMIM]^+$ cation were found to be good candidates for the dissolution of cellulose and hemicellulose. This was further confirmed by the measurement of solid-liquid equilibria with cellulose and hemicellulose. The regenerated cellulose powder was then characterised by Fourier transform spectroscopy(FTIR), X-ray diffraction (XRD) and Thermal gravimetric analysis (TGA).



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Cellulose/hemicellulose; ionic liquids; COSMO-RS; interaction energy; NBO analysis; molecular dynamics

1. Introduction

Over the past few decades, our society has facing many challenges due to the high consumption rate of fossil fuels. This has made a higher impact on the current environmental and economic processes for the production of energy and chemicals. Regrettably, along with the benefits, the process has led to an enhanced production of carbon dioxide in the environment [1,2]. Therefore, in recent years lignocellulosic biomass is recognised as a suitable feedstock for the production of fuels and chemicals [3–5]. Lignocellulosic biomass is composed of cellulose, hemicellulose and lignin. It is a known fact that in some cases cellulose is only a plurality of the total available biomass, rather than the primary component and upon hydrolysis which can produces glucose. The other component namely hemicellulose being the most abundant polymer in biomass produces pentose and hexose sugars [4]. In many plants, xylans are heteropolysaccharides with backbone chains of (1,4)-linked β-D-xylopyranose units. Among the key components of lignocellulosic biomass, hemicellulose is thermo-chemically sensitive [6]. The

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remaining moiety namely lignin is one of the most abundant polymers and a renewable bio-resource with unique functionalities. It is an asymmetrical, cross-linked and possesses a polyphenolic polymer which binds the wood compounds together. It is relatively aromatic while at the same time hydrophobic in nature. Lignin comprises of three phenylpropanoid monomers which are coniferyl alcohol, sinapyl alcohol and p-coumaryl alcohols [7].

One of the most important drawbacks is that the direct conversion of cellulose and hemicellulose into value-added chemicals is not possible [8,9]. Thus, it requires a suitable solvent for the dissolution of cellulose and hemicellulose without necessitating the pretreatment step [10]. The dissolution of cellulose and hemicellulose is a demanding task since water and other organic solvents are not able to dissolve these polymers at relatively milder conditions. This is due to the complex structure and strong inter- and intra-molecular hydrogen bonding which is present in the plant wall [11,12]. Hence, an extensive research is undertaken to break these complex structures by using different pretreatment methods [13]. In 2002, Rogers and co-workers have utilised the ionic liquids (ILs) for the dissolution of cellulose [14]. These solvents provide a new pathway for the dissolution of biomass and its derivatives under milder operating conditions without any pretreatment [15–17]. However, the disadvantage of ILs is its high viscosities which bring operating difficulties during the dissolution process. Nevertheless, as compared with bis[trifluoromethylsulfonyl]imide, tetrafluoroborate and hexafluorophosphate-based ILs, the ILs with acetate, thiocyanate, ethyl sulfate and phosphate anions have lower viscosities thereby enabling their utilisation in various applications [18,19].

There seems a little argument on anions such as $[OAc]^-$, $[AcF_3]^-$, $[Me_2PO_4]^-$ and $[Cl]^-$, which are good hydrogen bond acceptors and are known to be very effective in dissolving biomass and its derived components [20–23]. While ILs containing the low basicity anions, namely dicyanamide and thiocyanate are not efficient in dissolving cellulose although they are known to dissolve monosaccharides [24,25]. ILs containing non-coordinating anions such as $[BF_4]^-$ and $[PF_6]^-$ are also unable to dissolve cellulose [14,26]. In summary, the *n*-alkyl-imidazolium-based cations with $[OAc]^-$, $[Me2PO4]^-$ and $[Cl]^-$ anions are the recommended ILs for the dissolution of biomass and its derivatives [20,27]. It was reported that the dissolution of biomass and its derived components tends to depend on the anion, whereas other researchers inferred that it is the combined action of both the anion and cation [9,28]. Kahlen *et al.*, Casas *et al.* and Liu *et al.* employed the activity coefficient model namely COSMO-RS to study the effect of anions and cations on the solubility of cellulose and lignin [12,23,26,29]. In their study, they have considered cellotriose and cellotetraose as model cellulose molecules for COSMO-RS calculations.

Recently, Cheng *et al.* studied the impact of 1-ethyl-3-methylimidazolium acetate [EMIM][OAc] IL on the crystalline part of cellulose. 3–20 wt % of cellulose solution was treated with IL and mixture of IL and DMSO at 25 and 120 °C for 1 h. It was reported that the crystallinity of cellulose was found to increase with increasing cellulose loading (i.e. 3–20 wt %) [30]. Minnick *et al.* measured the solubility of cellulose (∼61% crystallinity) in 1-ethyl-3-methylimidazolium diethylphosphate [EMIM][DEP] IL at different temperatures ranging from 40 to 120 °C for 48 h. It was found that the solubility of cellulose reached its saturation point at 100 °C (19.8 wt %). Further increase in the temperature to 120 °C gave no increase in the solubility of cellulose [31]. Rabideau *et al.* applied molecular dynamics (MD) simulations to study the effect of different imidazolium-based cations for the solvation of cellulose with common anions namely: Cl^-, $[OAc]^-$ and $[DMP]^-$. From MD simulations, it was found that the cation interacts favourably with the nonpolar domains of cellulose through dispersion interactions, while the anions bounds at the polar domains of cellulose [32,33]. Lovell *et al.* reported the influence of cellulose on the ion diffusivity of [EMIM][OAc] through pulsed-field gradient Nuclear Magnetic Resonance (NMR). The self-diffusion coefficients of both $[EMIM]^+$ and $[OAc]^-$ ions were found to reduce with cellulose weight [34]. Furthermore, Uto *et al.* studied the crystal cellulose dissolution in imidazolium-based ILs by applying MD simulations. From their study, it was revealed that [AMIM]Cl and [EMIM]Cl are more powerful solvents for cellulose dissolution when compared to [BMIM][OAc] and [EMIM][OAc] [35]. Zhao *et al.* further studied the effect of anionic structures on cellulose dissolution. From their results, it was found that the interactions of the anions with cellulose decrease in the order: $Cl^- > [OAc]^- > [DEP]^- > [SCN]^- > [PF6]^-$ [36].

Furthermore, Guo *et al.* and Ding *et al.* studied the cellulose dissolution in ILs by both experimental and quantum chemical (QC) calculations [37,38]. Here, 1,4-O-dimethyl cellobiose and (1,4)-dimethoxy-β-D-glucose were considered as the cellulosic model. Furthermore, it was observed that the IL tends to form strong hydrogen bonds (H-bonds) with hydroxyl groups of cellulose molecule thereby causing the cellulose dissolution. In another work, Payal and Balasubramanian reported the cellulose dissolution in [Emim][OAc] IL and IL/water by an *ab initio* MD simulations [39]. They reported their prediction on two different types of cellulose conformers namely cellobiose-anti–anti and

cellobiose-anti-syn. Furthermore, Froschauer *et al.* have reported selective solvent for the xylan dissolution in [BMIM][Me_2PO_4] IL [40]. The phosphate anion was altered by replacing one oxygen atom with sulphur and selenium, respectively. This amendment decreases the hydrogen bond basicity of the IL. As a result, it precludes the dissolution of cellulose fibres, when compared to the less ordered xylan. After an extensive literature survey, the current work is another continuing theoretical contribution to elucidate the effect of individual ions on the solvation behaviour of cellobiose-anti–anti and cellobiose-anti-syn as model cellulose compounds. To the best of our knowledge, no data on the computational and experimental dissolution of hemicellulose in ILs is available.

Therefore, the present work reports the dissolution mechanism of cellulose and hemicellulose in ILs by quantifying the individual effect of anions and cations. Hence, a total of 1428 ILs have been screened in this work. An attempt has been made to elucidate the bond strength between the electron donor and acceptor orbitals. In this domain, our objective is to apply both QC and MD simulations for computing the IL-cellulose and IL-hemicellulose interactions. Here, the QC analysis included the COSMO-RS predictions, interaction energies and the natural bonding orbital (NBO) analysis. In addition, the effects of individual ions on different conformational structures of cellulose and hemicellulose are also explored. Furthermore, in order to confirm the findings of both COSMO-RS and QC calculations, experimental studies have also been performed for the selected ILs over the temperature ranges from 27 to 130 °C. The characterisation techniques namely Fourier transform spectroscopy (FTIR), X-ray diffraction (XRD) and Thermal gravimetric analysis (TGA) were also performed on the regenerated cellulose.

2. Computational details

2.1. COSMO-RS model and quantum chemical calculations

The structures of cellulose, hemicellulose, anions and cations were drawn with the MOLDEN freeware [41,42]. The structures of cellulose and hemicellulose were selected based on our previous works [4,43]. For the ease of computation, cellulose has been chosen as cellobiose-anti-syn and hemicellulose as comprised of an equimolar permutation of both glucose and xylose. The initial molecular geometries of all molecules were fully optimised by B3LYP/6-31G* *via Gaussian*03 package [44] and the COSMO files were generated by using final optimised geometries of investigated molecules. The detailed approach of COSMO-RS calculations is available in our previous works, hence not discussing here [4,24,42,43].

For the interaction energy study, the structures of cellulose, hemicellulose, anions, cations and ILs were drawn in MOLDEN by combining the molecules using a dummy atom. After the geometry optimisation, the final structure of optimised complex system appears without a dummy atom [4,45]. As the two molecules interact, a certain amount of energy gets reduced due to the solubility of two components. This decrease corresponds to the interaction energy between the molecules [45]. Mathematically, the interaction energy is computed by following expression [38,46]:

$$I.E\left(\text{kJ mol}^{-1}\right) = E_{AB} - (E_A + E_B) \tag{1}$$

Here, E_{AB} is total energy for the complex system AB (ion-cellulose or ion-hemicellulose) in kJ mol^{-1}. The complex system energy was corrected by the basis set superposition error using counterpoise method at the same level of theory [47]. E_A and E_B are the individual energies (kJ mol^{-1}) of A (anion or cation) and B (cellulose or hemicellulose) molecules, respectively. It should be noted that the negative energies are favourable because an extra amount of energy is required for disassembling the complex system [42,45].

NBO analysis has been carried at the B3LYP level of theory on 6-31G* basis set for a better understanding of the strength of intermolecular H-bonding interactions involved in the molecular systems of this study [46,48]. The stabilisation energy ($E^{(2)^*}$) associated with i (donor)—j (acceptor) delocalisation is estimated from the second-order perturbation approach as given in the following equation [37,49]:

$$E^{(2)*} = \Delta E_{(i,j)} = q_i \frac{F^2{}_{(i,j)}}{\varepsilon_j - \varepsilon_i} \tag{2}$$

where q_i is the donor orbital occupancy. ε_i and ε_j are the diagonal elements (orbital energies), and $F(i,j)$ is the off-diagonal NBO Fock matrix element.

2.2. Molecular dynamics simulations

The microscopic interactions between cellulose (i.e. cellobiose-anti-syn) and six different ILs ([EMIM][OAc], [BMIM][OAc], [EMIM][$MeSO_3$], [EMIM][$EtSO_4$], [EMIM][SCN] and [TMA][$MeSO_4$]) were performed by MD simulations using NAMD software [50]. In the entire simulations (COSMO-RS and QC), cellobiose molecule is considered as a cellulose model component. With the aim of avoiding the further confusions on

cellulose structure and due to the computational restrictions, we have adopted cellobiose molecule as cellulose model component to carry out the MD simulations. To hand, several literature reports are available for the solvation of cellulose using cellobiose and glucose as a cellulose model molecules [13,39,51]. It should be noted that the ILs were selected based on QC calculations and experimental evidence. For all the molecules, the CHARMM force field parameters were employed and presented in the supporting information (Tables S1–S8) along with their structures. The CHARMM force fields were well defined for carbohydrates and ILs [52–54] and successfully implemented for cellulose-IL systems [54,55]. The CHARMM force field parameters for cellulose (two-order degree of polymerisation) and $[BMIM]^+$ were taken from Guvench *et al.* [53] and Gross *et al.* [54]. The force field parameters for other cations ($[EMIM]^+$ and $[TMA]^+$) and anions ($[OAc]^-$, $[MeSO_3]^-$, $[EtSO_4]^-$, $[SCN]^-$ and $[MeSO_4]^-$) were developed with the existing CHARMM parameters [56]. The developed force field parameters were further validated by computing the density of ILs at 300 K and 1 atm pressure. For measuring the density of ILs, 200 ionic pairs of IL molecule were taken. The experimental and MD simulated densities gave an excellent agreement between them and are given in Table S9.

All the MD simulations were performed in Langevin thermostat and Nosé–Hoover Langevin barostat at constant temperature (300 K) and pressure (1 atm) [57]. A damping coefficient of 1 ps^{-1} was used for temperature control. The Nosé–Hoover barostat with an oscillation period of 100 fs and a damping factor of 50 fs was used for pressure control [58]. As per the NAMD user guidelines [59], it is recommended that the product of full electrostatic frequency and time step does not exceed 4 fs unless rigid bonds (i.e. SHAKE) should be used [50]. In the present study, the full electrostatic frequency of 2.0 is used and the product of full electrostatic frequency and time step (1 fs) is 2 fs. The current work has adopted SHAKE algorithm with a tolerance of 10^{-5} so as to constrain the lengths of covalent bonds which are connected to hydrogen [60]. The particle mesh Ewald (PME) method was applied with an accuracy of 10^{-6} (PME tolerance) for long-range electrostatic interactions at a cut-off distance of 15 Å [61]. The initial configuration of cellulose/ILs pairs was prepared by PACKMOL [62] in a cubic box containing 200 ion pairs of ILs and 10 cellulose molecules. The size of the cubic box was typically larger than the actual box size which makes the packing easier [55,63]. Three different initial configurations were generated in order to get the starting geometry.

Initially, the simulation was carried for 0.5 ns for energy minimisation. The NAMD sample script for energy minimisation was provided in the supporting information and the energy minimisation plot is shown in Figure S1. After energy minimisation, the system was then gradually heated to 300 K for 0.5 ns. At target temperature, the simulated system was equilibrated for 5 ns with isothermal-isobaric ensemble (NPT) in order to converge the system to its experimental condition. This was monitored by measuring the density or the volume of the system. Furthermore, as per literature data, long-time dynamics are necessary for the simulation system containing crystalline cellulose with more than 10 degree of polymerisation (i.e. consisting of at least 10 cellulose chains) which is inherently proven from previous work [28,32,64]. The simulations with smaller cellulose model component molecules such as glucose, cellobiose and cellopentaose require 5–30 ns of production runs to diffuse significantly [13,39,48,65]. Therefore, the production run has been attempted for 60 ns with canonical (NVT) ensemble which does not show any appreciable variation with respect to non-bonded energies (see Figure S2). At every 5 ps, the trajectory data was saved for structural analysis such as radial distribution functions (RDF) and the non-bonded interaction energy calculations. All the MD computational trajectories were then visualised with the VMD software [66]. The interaction energy between the cellulose and ILs was calculated per mole of cellulose. Furthermore, to characterise the stability of the cellulose molecule in ILs, we have calculated the root mean square deviation (RMSD) values in molecular positions of the cellulose as shown in Figure S3. The RMSD curve in both ILs [EMIM][OAc] and [BMIM][OAc] is fluctuating in nature indicating that the molecular positions changed sharply during the initial part of simulation (<15 ns) after which steady-state diffusion of cellulose molecule occurs.

3. Materials and methods

3.1. Materials

Microcrystalline cellulose (MCC) powder was purchased from Merck, India. The degree of polymerisation of MCC powder was ≤400. Xylan (hemicellulose) from beech wood was purchased from Sigma-Aldrich, Germany. The ILs, [EMIM][OAc], [BMIM][OAc], $[EMIM][MeSO_3]$, $[EMIM][EtSO_4]$, [EMIM][SCN] and $[TMA][MeSO_4]$ were purchased from Sigma-Aldrich, Germany. All the chemicals were used without any further purification. Table 1 summarises the sample description of cellulose, hemicellulose and ILs along with their moisture content.

Table 1. Chemicals used in dissolution experiments.

Sl. No.	Compound name	Purity (%)	CAS No	Water content (ppm)	Supplier	Product form
	Microcrystalline cellulose	≥97	9004-34-6	NM	Merck	India
2.	Xylan from beech wood	≥90	9014-63-5	NM	Sigma-Aldrich	Germany
3.	[EMIM][OAc]	≥95	143,314-17-4	11 800	Sigma-Aldrich	Germany
4.	[BMIM][OAc]	≥95	284,049-75-8	9650	Sigma-Aldrich	Germany
5.	[EMIM][$MeSO_3$]	≥95	145,022-45-3	3600	Sigma-Aldrich	Germany
6.	[EMIM][$EtSO_4$]	≥95	342,573-75-5	3000	Sigma-Aldrich	Germany
7.	[EMIM][SCN]	≥95	331,717-63-6	7400	Sigma-Aldrich	Germany
8.	[TMA][$MeSO_4$]	≥95	29,463-06-7	13 100	Sigma-Aldrich	Germany

The moisture content of ILs was determined by a Karl Fischer titrator (Metrohm, 787 KF Titrino, Switzerland). Cellulose and xylan were dried at 50 °C for 24 h before the experiments.

3.2. Experimental procedure

In a typical dissolution experiment, the ILs [EMIM][OAc], [BMIM][OAc], [EMIM][$MeSO_3$], [EMIM][$EtSO_4$], [EMIM][SCN] and [TMA][$MeSO_4$] were added into a 25 ml conical flask containing 6 g of IL. It was then immersed in an oil bath. The temperature of the system was maintained separately at individual temperatures ranging from 27 to 130 °C (i.e. 27, 50, 70, 90, 110 and 130 °C) with a precision of ±0.1 K (TARSONS SPINOT-magnetic stirrer and hot plate-DIGITAL, MC 02, India). The inside temperature of the flask was measured with a thermometer and maintained at the desired temperature. After the desired temperature was achieved, 0.3 g of cellulose (for cellulose dissolution) and 0.5 g hemicellulose (for hemicellulose dissolution) were added. The heated mixture was magnetically stirred for different individual time intervals, i.e. 0.5, 1 and 2 h to know the effect of time on the solubility study. After completion of dissolution time, cellulose and hemicellulose solutions were cooled to room temperature and then regenerated by adding distilled water into the solution with vigorous stirring for 30 min. The cellulose and hemicellulose flocs were then collected by a vacuum filtration and dried at 50 °C for 48 h. All the experiments were performed in triplicate and average values are reported here. Furthermore, the dried cellulose samples were characterised by FTIR, XRD and TGA. The detailed descriptions of FTIR and XRD characterisations are given in our previous studies [2,3,67]. The per cent of solubility was calculated according to the following equation:

$$\text{Cellulose or hemicellulose solubility (\%)} = \frac{m_o - m_f}{m_o} \times 100 \quad (3)$$

where m_o is the initial mass of the cellulose or hemicellulose added and m_f the final mass of the recovered cellulose and hemicellulose, respectively.

4. Results and discussion

4.1. Screening of ionic liquids by COSMO-RS model

As per the standard definition, ILs can be freely combined from organic cations and organic/inorganic anions, thereby forming numerous candidates for the dissolution of biomass. COSMO-RS is a valuable tool for a predefined screening using infinite dilution activity coefficient (IDAC) predictions. In our study, 42 cations and 34 anions comprising of 1428 combinations were ranked based on their IDAC values (Figures 1 and 2). The IDAC value refers to the value of solute (cellulose/hemicelluloses) in IL (solvent), where the reciprocal gives the extent of solubility of the solute in IL [24,68]. The structures and properties of 42 cations and 34 anions are provided in the supplementary information (Tables S10 and S11). At this point, it should be noted that all the investigated cation–anion combinations were not liquids at room temperature and hence, a further selection based on the liquid state was carried out comprehensively. Such combination with a predictive model with IL melting temperatures has been proposed by Krossing *et al.* [69] for sorting out the solid cation–anion combinations. It should be noted that without a screening of potential ILs, experimental studies become expensive and time consuming.

The IDAC values are often used as a quantitative descriptor for the dissolution power of a solvent. Cations and anions were sorted according to their dissolving capacity and arranged in such way that ILs with high dissolving power (i.e. $\ln \gamma^{\infty}_{\text{cellulose/hemicellulose}} << 0$) are situated in the midline to bottom portion of Figures 1 and 2, and the weaker ones (i.e. positive values of $\ln\gamma^{\infty}_{\text{cellulose/hemicellulose}}$) are situated at the top portion of Figures 1 and 2. As can be seen from Figures 1 and 2, anions such as acetate, chloride, nitrate, bromide, iodide,

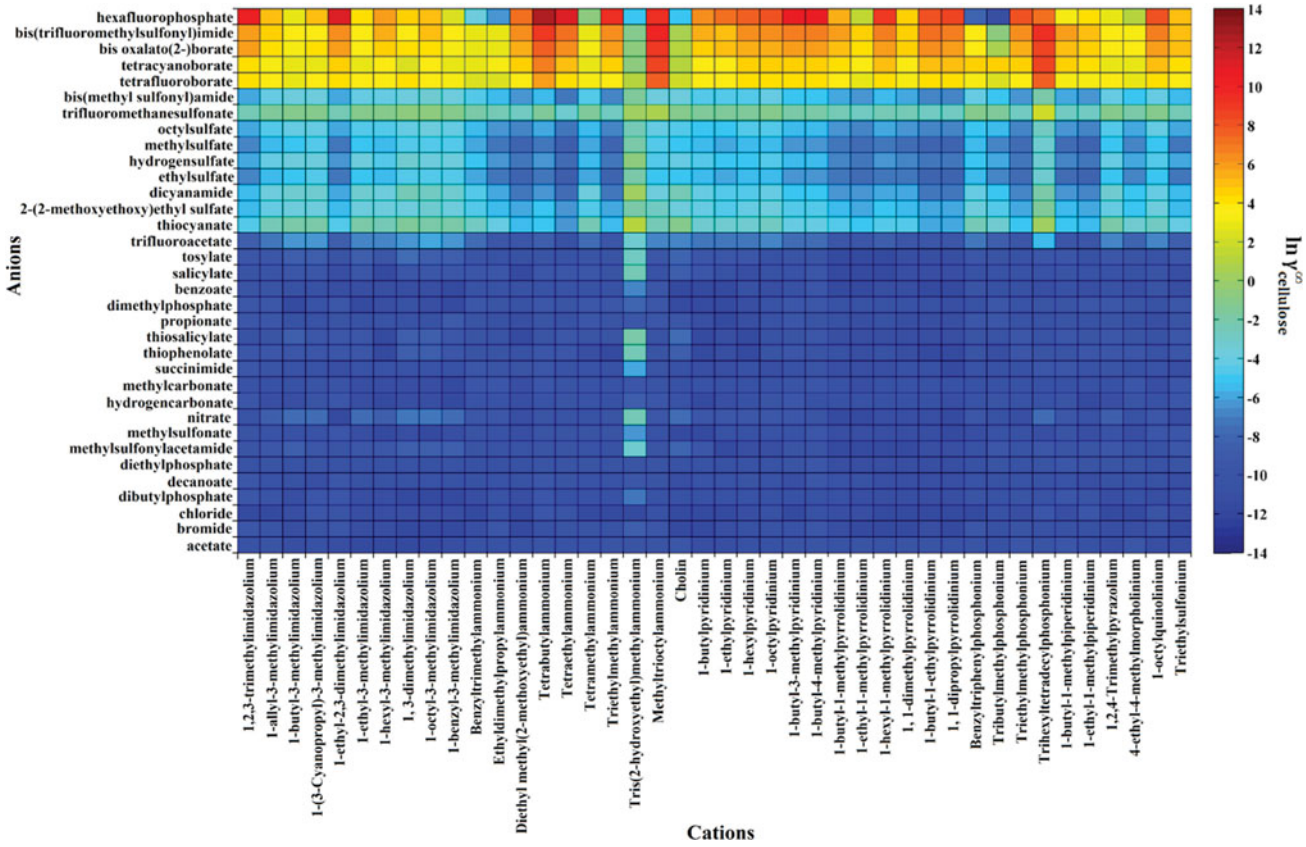

Figure 1. (Colour online) Graphical representation of the infinite dilution activity coefficients of cellulose (cellobiose-anti-syn) in 1428 different ILs at room temperature by COSMO-RS model.

dimethyl phosphate, hydrogen carbonate and methyl carbonate are predicted to dissolve the cellulose and hemicellulose molecules more efficiently with most cations. However, in combination with anions such as $[PF_6]^-$, $[BF_4]^-$, tetracyanoborate and bis(trifluoromethylsulfonyl)imide have very high values of IDAC. The imidazolium-based and other cations like ammonium and pyrrolidinium remarkably favourable for the solubility predictions (Figures 1 and 2). This is due to the fact that the solvent molecules form strong H-bonds with cellulose and hemicellulose, which alters the extensive H-bond network present within the solute molecules [70]. This phenomenon was elucidated by Kahlen *et al.* that the energy of the H-bonds of cellulose is up to 25 kJ mol^{-1} [71], thus the interactions between cellulose and solvent need to go beyond this energy in order to result in cellulose dissolution. Therefore, the ions of IL should obey the following successive criteria: (1) either of the ions should be highly polar or in other words act as a good hydrogen bond acceptor or donor, and (b) the remaining ion to be slightly polar [29].

According to the above assumptions, the ions such as 1-ethyl-3-methylimidazolium ($[EMIM]^+$) cation (less polar) and acetate ($[OAc]^-$) anion (highly polar) (Figure 3) are known to be the recommended IL. The interaction between a polar anion and the hydroxyl groups of cellulose or hemicellulose molecules are energetically much stronger than the interaction between anion and cation of an IL, resulting in a high solvation of cellulose and hemicellulose [70]. Kahlen *et al.* performed a comprehensive COSMO-RS screening study for cellulose dissolution in 2272 ILs resulting from the combinations of 32 anions and 71 cations [29]. Furthermore, Li *et al.* also screened 357 IL pairs for cellulose solubility [12]. Kahlen *et al.*and Liu *et al.* used cellotriose and cellotetraose as cellulose model compounds, whereas in our case cellobiose is used as a cellulose model component. However, the results of COSMO-RS screening study in the current work are in line with the reported findings of both Kahlen *et al.* and Li *et al.*. Overall with increase in the cation chain length the IDAC was found to decrease. On the other hand, the sigma profiles of glucose, cellobiose, cellotriose and cellopentaose are presented in Figure S4. From Figure S4, it is seen that the peak intensity of sigma profiles increases with cellulose chain length irrespective of the magnitude of screening charge densities. The sigma profiles of glucose, cellobiose, cellotriose and cellopentaose show an increasing trend

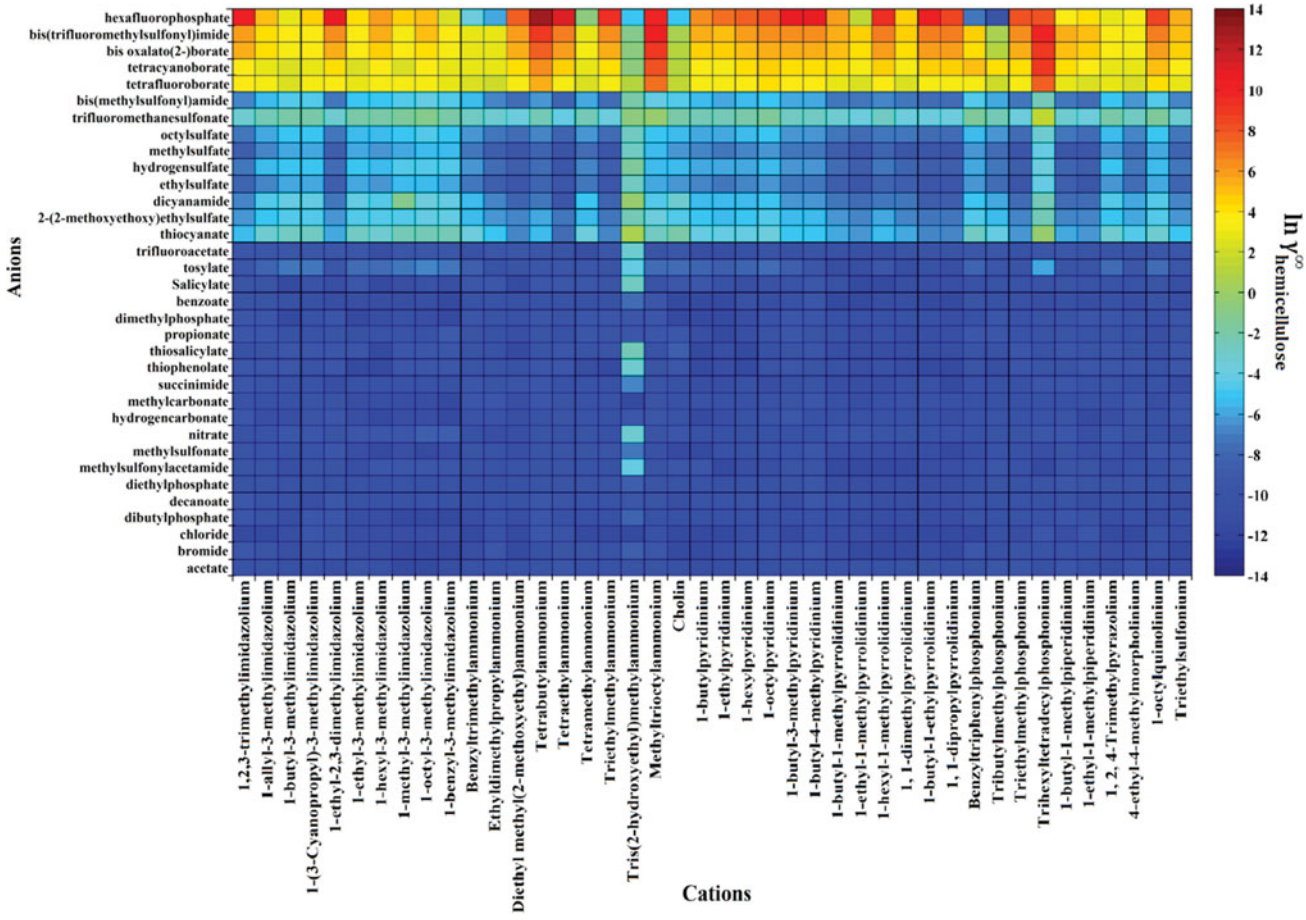

Figure 2. (Colour online) Graphical representation of the infinite dilution activity coefficients of hemicellulose (xylose + glucose) in 1428 different ILs at room temperature by COSMO-RS model.

in both H-bond donor and acceptor zone with increasing glucan rings implying the fact that the polarity of cellulose increases with the number of glucan rings.

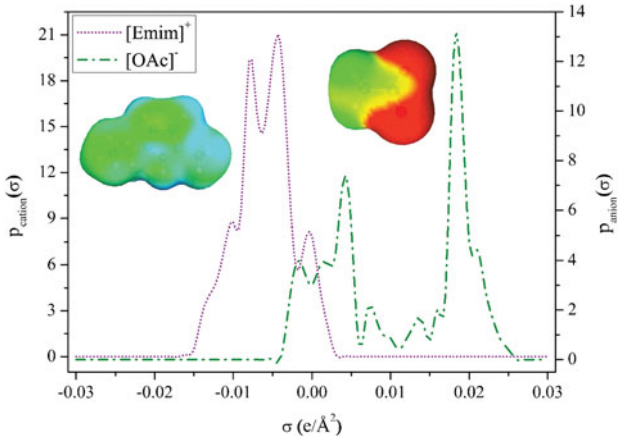

Figure 3. (Colour online) Sigma profiles and COSMO-cavities of cation $[EMIM]^+$ and of the anion $[OAc]^-$. Here the extent of screening charge varies from −0.03 e/Å^2 (red) to +0.03 e/Å^2 (blue) within the molecule, while the intermediate region is represented by green and yellow colour

4.2. Interaction energy and NBO analysis

4.2.1. Effect of anions on cellulose and hemicellulose dissolutions

The objective for performing such a study is locating the favourable site within the anion for the incoming cellulose/hemicellulose molecule. For these molecules, trial QC calculations were performed for different sites of cellobiose with acetate anion (Figure S5). From Figure S5, it is clear that the most stable system for acetate anion to attach with the cellobiose molecule is around O3 and O4 atoms (Figure 4). This is particularly evident from Figure S5(a,b) where the interaction energies are the highest among all the conformers. Similar calculations were also performed for hemicellulose and found that the favourable sites for acetate anion to attach hemicellulose molecule were around O3, O4 and O8 atoms, respectively (Figure 4). Thereafter, the anions were placed around these sites. The interaction energy is broken down into anion–cellulose/hemicellulose and cation–cellulose/hemicellulose pairs to understand the effect of individual ions on the solvation of cellulose

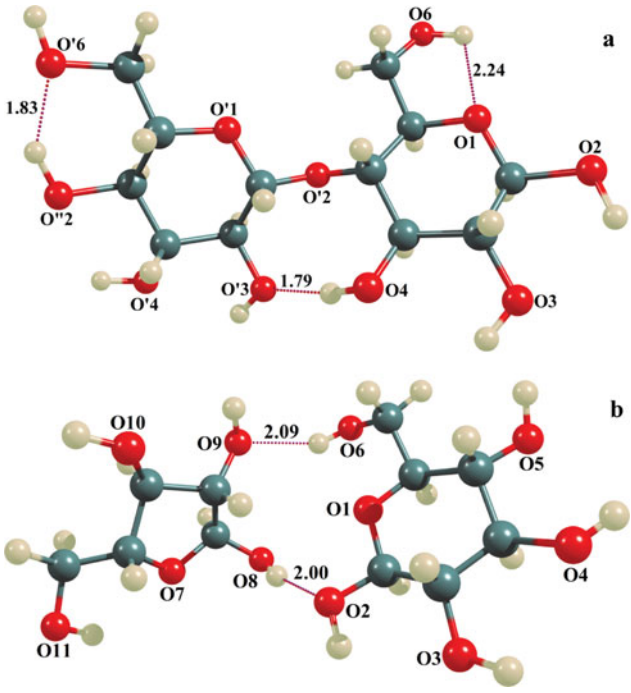

Figure 4. Optimised geometries of (a) cellulose (cellobiose-anti-syn) and (b) hemicellulose (xylose + glucose) with corresponding atom numbers. The bond lengths are in Angstrom (Å).

and hemicellulose [37,38]. This has been primarily performed to screen the potential anions and cations. This is required since the overall interaction energy for IL-cellulose/hemicellulose system is significant. However, it is rather cumbersome to understand the effect of each ion on the cellulose/hemicellulose molecules due to the nature of its binding which is short range in QC. Therefore, the interaction energy is broken down into ion-cellulose/hemicellulose pairs.

The following anions were chosen: $[OAc]^-$, $[MC]^-$, $[AcF_3]^-$, $[Cl]^-$, $[HC]^-$, $[Me_2PO_4]^-$, $[NO_3]^-$, $[MeSO_3]^-$, $[EtSO_4]^-$, $[MeSO_4]^-$, $[MSF_3]^-$, $[BF_4]^-$ $[PF_6]^-$ and $[SCN]^-$ to investigate the interaction energy and NBO analysis. The interaction energies of cellulose and hemicellulose with the 14 anions are presented in Table 2. From the table, it can be seen that the $[OAc]^-$ gave the highest interaction energy with cellulose (−227.02 kJ mol^{-1}) and hemicellulose (−222.28 kJ mol^{-1}). Based on their interaction energies from Table 2, the ILs with $[OAc]^-$, $[HC]^-$, $[MC]^-$, $[Cl]^-$ and $[AcF_3]^-$ anions are probable candidates for the dissolution of cellulose and hemicellulose. In contrast, the anions $[BF_4]^-$, $[MSF_3]^-$, $[SCN]^-$ and $[PF_6]^-$ exhibit lower interaction energies, which result in their inability to disturb the intramolecular H-bonds.

A further insight is depicted in Figures 5 and 6. From the observation of Figures 5(a) and 6(a), the complexes namely cellulose-$[OAc]^-$ and hemicellulose-$[OAc]^-$ are energetically more stable than other complexes. The intermolecular distances (O-H…O) of cellulose and hemicellulose molecules with $[OAc]^-$ anion were less than 1.8 Å indicating strong H-bonding. The oxygen atoms of $[OAc]^-$ anion had higher negative partial charge, which gave itself more strength to interact with the other molecule. A similar trend was observed for cellulose/hemicellulose with $[MC]^-$ (Figures 5(d) and 6(d)), where the intermolecular distances were less than 1.9 Å. The interaction energy between cellulose-$[HC]^-$ and hemicellulose-$[HC]^-$ was found to be lower than the $[OAc]^-$ and $[MC]^-$ anion interaction energies. The strength of the H-bonds between the anion ($[AcF_3]^-$ and $[HC]^-$) and hydroxyl hydrogen atoms of cellulose/hemicellulose was relatively weaker than $[OAc]^-$ and $[MC]^-$ anions strengths (Figures 5(b,c) and 6(b,c). Similarly, the anions such as $[Me_2PO_4]^-$, $[Cl]^-$, $[NO_3]^-$ $[MeSO_3]^-$, $[EtSO_4]^-$ and $[MeSO_4]^-$ also interacted with cellulose and hemicellulose *via* H-bonds. The bond distances and partial charges of these anions with

Table 2. Interaction energies of cellulose (cellobiose-anti-syn) and hemicellulose (xylose + glucose) with anions.

		Interaction energy (kJ mol^{-1})	
Entry	Name of the anion	Cellulose	Hemicellulose
1	Acetate $[OAc]^-$	− 227.02	− 222.28
2.	Methylcarbonate $[MC]^-$	− 212.80	− 205.48
3.	Trifluoroacetate $[AcF_3]^-$	− 192.22	− 182.30
4.	Chloride $[Cl]^-$	− 187.52	− 178.03
5.	Hydrogencarbonate $[HC]^-$	− 201.57	− 207.58
6.	Dimethylphosphate $[Me_2PO_4]^-$	− 185.52	− 198.61
7.	Nitrate $[NO_3]^-$	− 182.77	− 166.81
8.	Methanesulfonate $[MeSO_3]^-$	− 171.63	-
9.	Methylsulfate $[MeSO_4]^-$	− 169.55	− 161.31
10.	Ethyl sulfate $[EtSO_4]^-$	− 161.29	-
11.	Tetrafluoroborate $[BF_4]^-$	− 157.61	− 144.01
12.	Trifluoromethanesulfonate $[MSF_3]^-$	− 147.03	− 135.25
13.	Thiocyanate $[SCN]^-$	− 137.25	− 137.55
14.	Hexafluorophosphate $[PF_6]^-$	− 130.98	− 115.24

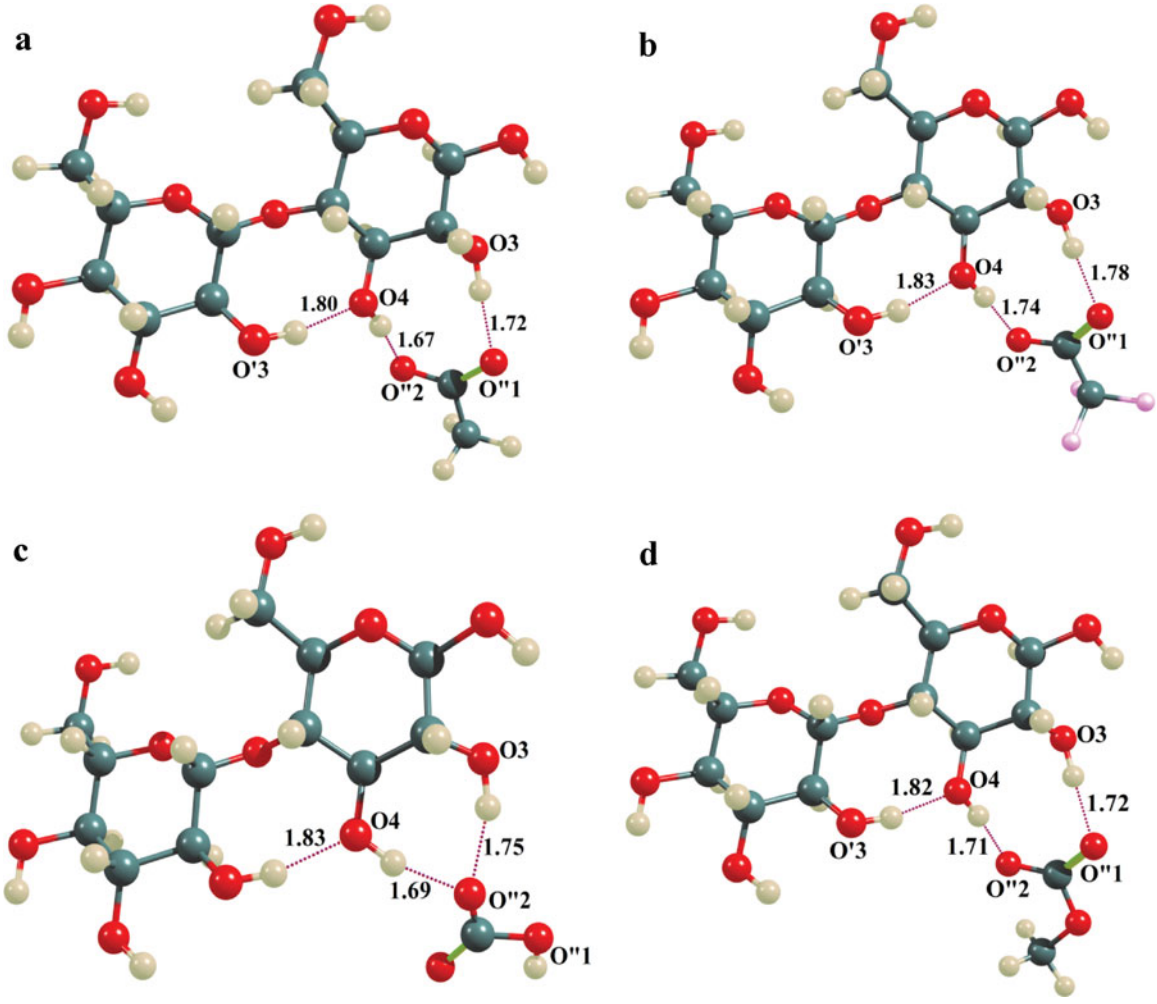

Figure 5. Optimised geometries for cellulose-anions (a) $[OAc]^-$, (b) $[AcF_3]^-$, (c) $[HC]^-$ and (d) $[MC]^-$. The H-bonds are indicated by dotted lines, the bond lengths are in Angstrom (Å) and given with corresponding cellulose atom numbers (Figure 4(a)).

cellulose and hemicellulose are reported in Tables S12 and S13.

Furthermore, the strength of the bond between electron donor–acceptor interactions was explained by NBO analysis, which corresponds to the second-order perturbation approach. The bond energies $E^{(2)^*}$ of electron donor–acceptor are reported in Table 3. In the complex system (Figures 5 and 6), the anion moieties are the electron donors and cellulose/hemicellulose moieties are the electron acceptors. In all the four complexes (Table 3), the lone pair electrons of the anions (oxygen atoms) interact with the antibonding orbitals (BD^*) of cellulose and hemicellulose (hydroxyl proton). These electron donor–acceptor orbitals are responsible for the formation of strong H-bonds between them which is indicated in the bond lengths as discussed earlier. Among all the complexes, the $E^{(2)^*}$ values of electron donor–acceptor orbitals which involve the acceptor orbital $BD^*(1)O4–H$ for cellulose; and $BD^*(1)O3–H$ for hemicellulose are relatively stronger than that of other donor–acceptor orbitals. It is a known fact that a lone pair donor → antibonding acceptor orbital interaction will weaken the bond associated with the antibonding orbital [49]. Among all the configurations of cellulose and hemicellulose with anions, $[OAc]^-$ anion has displayed a higher H-bond energy than that of other donors (Table 3). Because of the phenomena, $[OAc]^-$ had more interaction energy with cellulose and hemicellulose. Therefore, it can be concluded that the $[OAc]^-$ anion-based ILs have excellent dissolution capacity for the biomass and its derivatives as well.

4.2.2. Effect of cations on cellulose and hemicellulose dissolution

In a similar manner, QC calculations were performed for cellulose and hemicellulose with $[EMIM]^+$ cation. It was

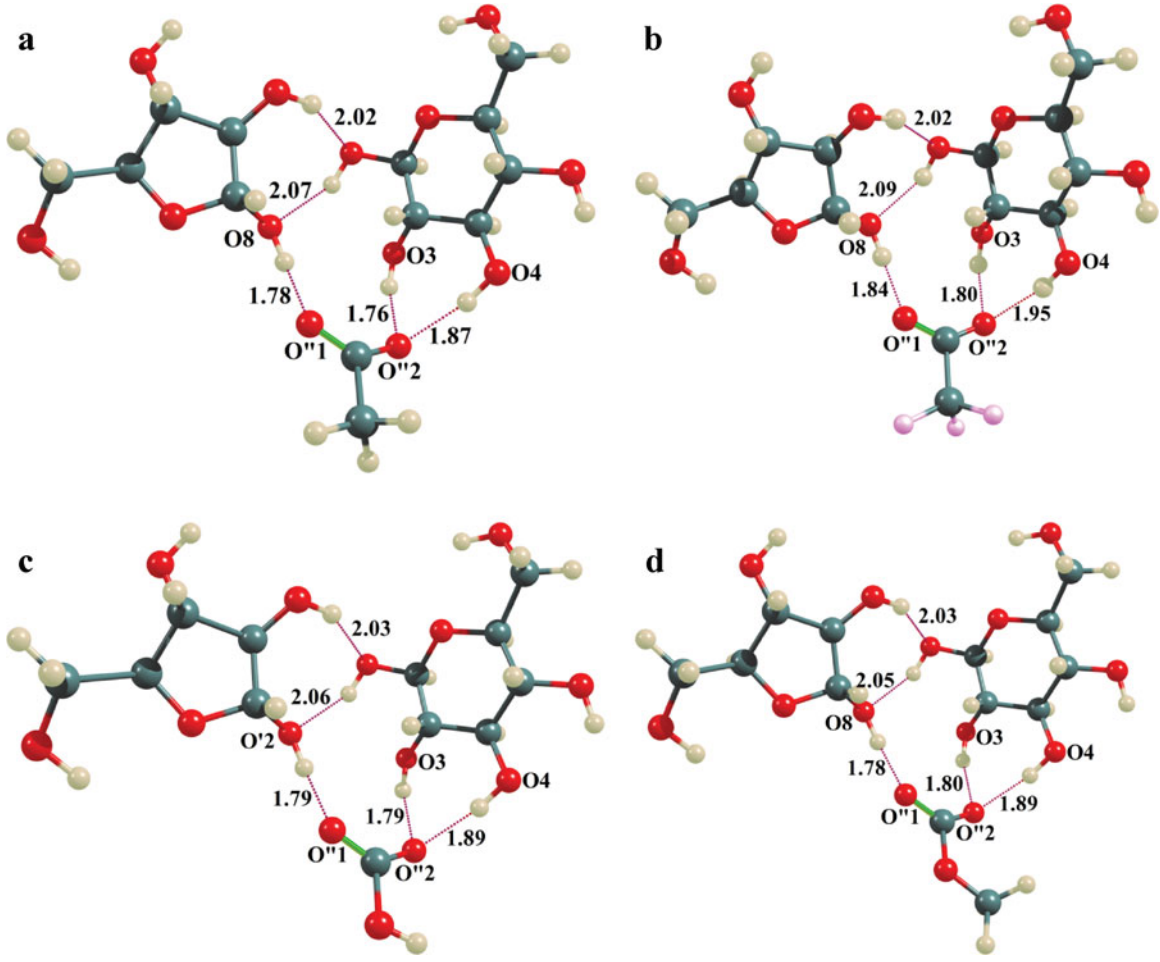

Figure 6. Optimised geometries for hemicellulose–anions (a) $[OAc]^-$, (b) $[AcF_3]^-$, (c) $[HC]^-$ and (d) $[MC]^-$. The H-bonds are indicated by dotted lines, the bond lengths are in Angstrom (Å) and given with corresponding hemicellulose atom numbers (Figure 4(b)).

Table 3. The electron donor and acceptor orbitals with their corresponding second-order interaction energies $E^{(2)*}$ (NBO analysis).

		$E^{(2)*}$ (kJ mol^{-1})			
Donor	Acceptor	$[OAc]^-$	$[AcF_3]^-$	$[MC]^-$	$[HC]^-$
Cellulose (cellobiose-anti-syn)					
LP (1) O″1	BD*(1) O3 – H	31.76	32.51	40.96	–
LP (2) O″1	BD*(1) O3 – H	113.80	80.04	101.09	–
LP (1) O″2	BD*(1) O4 – H	35.19	37.61	38.83	49.62
LP (2) O″2	BD*(1) O4 – H	140.46	95.27	108.49	105.31
LP (3) O″2	BD*(1) O4 – H	11.67	7.53	– – –	6.07
LP (1) O″2	BD*(1) O3 – H	–	– – –	–	84.18
LP (2) O″2	BD*(1) O3 – H	–	–	–	4.06
LP (3) O″2	BD*(1) O3 – H	–	–	–	25.90
Hemicellulose (xylose + glucose)					
LP (1) O″2	BD*(1) O3 – H	20.63	25.61	24.18	23.56
LP (2) O″2	BD*(1) O3 – H	103.55	77.40	81.84	84.85
LP (3) O″2	BD*(1) O3 – H	–	0.38	0.38	0.46
LP (1) O″2	BD*(1) O4 – H	39.33	31.46	35.27	35.31
LP (2) O″2	BD*(1) O4 – H	4.81	2.05	4.27	5.06
LP (3) O″2	BD*(1) O4 – H	37.66	26.11	36.11	37.07
LP (1) O″1	BD*(1) O8 – H	43.60	41.88	52.55	51.42
LP (2) O″1	BD*(1) O8 – H	78.78	50.75	61.63	60.08

Table 4. Interaction energies of cellulose (cellobiose-anti-syn) and hemicellulose (xylose + glucose) with cations.

Entry	Name of the cation	Interaction energy (kJ mol^{-1})	
		Cellulose	Hemicellulose
1.	1-Ethyl-3-methylimidazolium [EMIM]$^+$	− 116.26	−113.24
2.	1-Butyl-3-methylimidazolium [BMIM]$^+$	− 111.34	−103.68
3.	1-Allyl-3-methylimidazolium [AMIM]$^+$	− 109.35	−81.70
4.	Tetramethylammonium [TMeA]$^+$	− 107.31	−38.81
5.	Triethylmethylammonium [TEMA]$^+$	− 94.50	−73.99
6.	Tris(2-hydroxymethyl)methylammonium [TMA]$^+$	− 66.04	–
7.	1-Ethylpyridinium [EPy]$^+$	− 106.22	−74.60
8.	1-Butyl-3-methylpyridinium [B3MPy]$^+$	− 106.45	−78.98
9.	1-Butyl-1-methylpyrrolidinium [BMPyr]$^+$	− 87.25	−39.67
10.	Triethylsulfonium [TES]$^+$	− 93.61	−68.92

Table 5. The electron donor and acceptor orbitals with their corresponding second-order interaction energies $E^{(2)*}$ (NBO analysis).

Donor	Acceptor	$E^{(2)*}$ (kJ mol^{-1})	
		[EMIM]$^+$	[BMIM]$^+$
Cellulose (cellobiose-anti-syn)			
LP (1) O2	BD*(1) C″1 – H	3.81	3.60
LP (2) O2	BD*(1) C″1 – H	13.18	12.55
Hemicellulose (xylose+glucose)			
LP (1) O10	BD*(1) C″1 – H	34.14	32.10

noticed that the encouraging sites in the cellulose and hemicellulose molecules for the [EMIM]$^+$ cation results around the O2 and O10 atoms, respectively (Figure 4) and all the cations were placed at the selected sites of solutes. Other cations such as [BMIM]$^+$, [AMIM]$^+$, [TMeA]$^+$, [TEMA]$^+$, [TMA]$^+$ [EPy]$^+$, [B3MPy]$^+$, [BMPyr]$^+$ and [TES]$^+$ were also explored. The interaction energies of cellulose and hemicellulose with these 10 cations are presented in Table 4. Among all the cations, [EMIM]$^+$ has the strongest interaction with cellulose (−116.26 kJ mol^{-1}) and hemicellulose (−113.24 kJ mol^{-1}). On the other hand, imidazolium-based cations exhibit high interaction energies. The interaction strengths of cation for cellulose were found to follow: [EMIM]$^+$ > [BMIM]$^+$ > [AMIM]$^+$ > [TMeA]$^+$ > [EPy]$^+$ > [B3MPy]$^+$ > [TEMA]$^+$ > [TES]$^+$ > [BMPyr]$^+$ > [TMA]$^+$ and for hemicellulose: [EMIM]$^+$ > [BMIM]$^+$ > [AMIM]$^+$ > [B3MPy]$^+$ > [EPy]$^+$ > [TEMA]$^+$ > [TES]$^+$ > [BMPyr]$^+$ > [TMeA]$^+$.

Apart from interaction energy, the NBO analysis for [EMIM]$^+$ and [BMIM]$^+$ cations with cellulose and hemicellulose are presented in Table 5. The optimised geometry configurations and H-bonding distance between the hydroxyl groups of cellulose and hemicellulose with cations are presented in Figure 7. From the figure, the cation moieties act as electron acceptors while the cellulose/hemicellulose moieties as electron donors. In both complexes, the lone pair electrons of hydroxyl oxygen atoms of the cellulose (O2) and hemicellulose (O10) interact with the antibonding orbitals of cations (C″1–H). The bond length (C–H …O) of cellulose and hemicellulose with cations varies from 2.12 to 2.33 Å, which is more than anion bond lengths. These results confirm the reason; the anions have high interactions with cellulose and hemicellulose than cations. Furthermore, the bond energies of [EMIM]$^+$ cation are relatively higher than that of [BMIM]$^+$ (see Table 5). As a result, [EMIM]$^+$ cation-based ILs are the better solvents for the dissolution of biomass. On comparing the anion interaction with cations, it was found that the O–H …O bond is stronger than that of C–H …O bond. Therefore, the anionic part of the IL plays a predominating role in the dissolution process. Keeping the advantages in mind the IL: [EMIM][OAc] was chosen to dissolve cellulose/hemicellulose experimentally so as to validate our predictions and comparison was made with respect to cellulose dissolution in other ILs. Hemicellulose was dissolved only in [EMIM][OAc] IL as our main target was on cellulose molecule owing to its highly crystalline structure.

4.3. Effect of anions and cations on different conformational structures of cellulose and hemicellulose

Furthermore, we have studied the effect of different anions ([OAc]$^-$, [AcF$_3$]$^-$, [MC]$^-$) and cations ([EMIM]$^+$ [BMIM]$^+$) on the conformational structures of cellulose and hemicellulose. The considered conformational structures for cellulose are cellobiose-anti-syn and cellobiose-anti–anti. The counterparts for hemicellulose are xylose + glucose and xylobiose, respectively (Figure S6). Table 6 reports the interaction energies of cellulose and hemicellulose with anions and cations for different conformers. From Table 6, it can be seen that the interaction energy

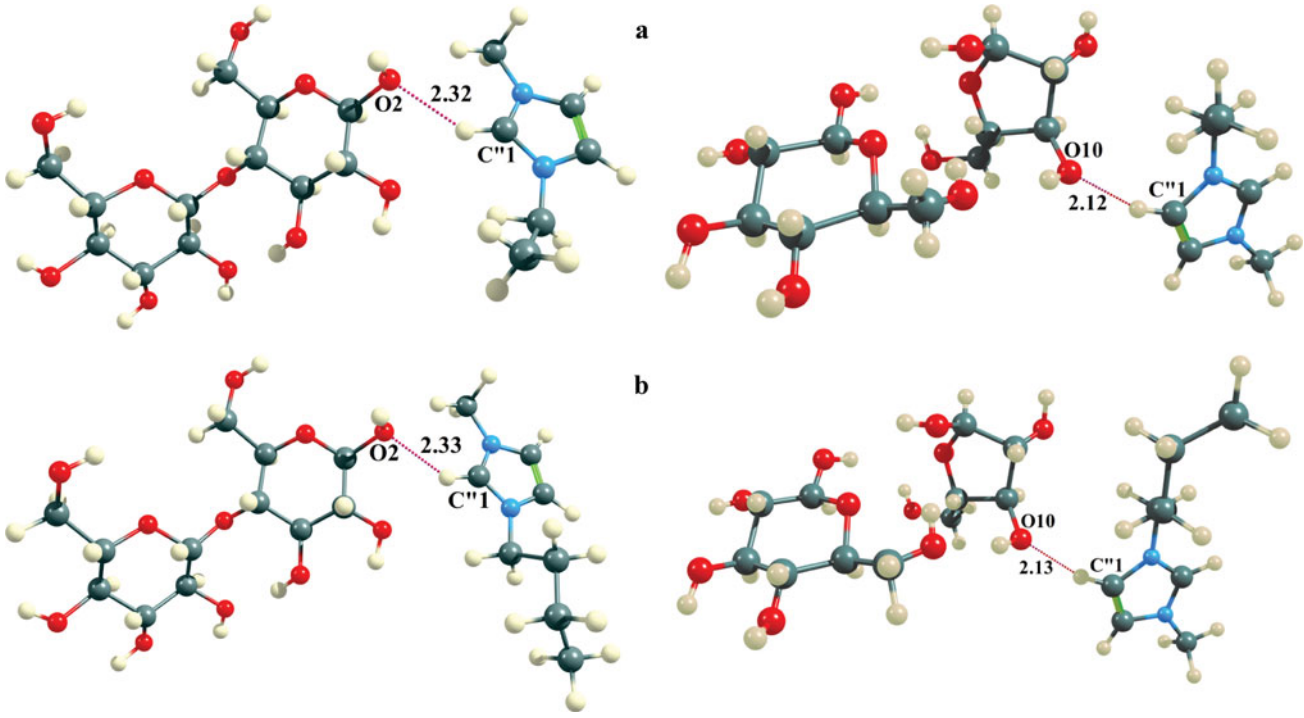

Figure 7. Optimised geometries for cellulose and hemicellulose with cations (a) $[EMIM]^+$ and (b) $[BMIM]^+$. The H-bonds are indicated by dotted lines, the bond lengths are in Angstrom (Å), given with corresponding atom numbers (Figure 4).

between cellobiose-(anti-syn)-anions/cations is higher than other conformers of cellulose, i.e. cellobiose-(anti–anti). The interatomic distance between the cellulose-anions (O–H…O) bond representing the cellobiose-(anti-syn)-anions is found to be in the range of 1.67–1.78 Å. On a similar line, the distances for cellobiose-(anti–anti)-anions are in the range of 1.71–1.89 Å (Figure S7). This eventually implies a higher strength of the H-bonds between the anions and hydroxyl hydrogen atoms of cellulose-anti-syn. Furthermore, the strength of the bond between the electron donor–acceptor interactions was elucidated by NBO analysis. From Tables S14 and S15, it was also found that the bond energies $E^{(2)*}$ of electron donor–acceptor were greater for 'cellobiose-anti-syn' system when compared with 'cellobiose-anti–anti' systems.

Table 6 also reports the interaction energies of hemicellulose-anions/cations with different conformers of hemicellulose. The conformer xylose + glucose exhibited higher interaction energy than xylobiose. The interaction energy deviation between the hemicellulose conformers was relatively higher compared to cellulose conformers. This higher interaction energy deviation resulted due to the difference in number of hydroxyl groups present in conformers. The intra-molecular H-bonds of xylose + glucose also get weakened by the formation of strong inter-molecular H-bonds with the anion. On the contrary, xylobiose retains its strong intra-molecular H-bond (Figure S8). Thus, the interaction energy was higher for xylose + glucose conformer. Furthermore, these results were confirmed by establishing the NBO analysis (Tables S16 and S17). In

Table 6. Interaction energies for the conformers of cellulose and hemicellulose with anions and cations.

		Cellulose I.E. (kJ mol^{-1})		Hemicellulose I.E. (kJ mol^{-1})	
Entry	Name of the anion/cation	Cellobiose-anti-syn	Cellobiose-anti–anti	Xylose + glucose	Xylobiose
1	$[OAc]^-$	− 227.02	− 218.23	− 222.28	− 190.62
2.	$[MC]^-$	− 212.80	− 208.57	− 205.48	− 178.07
3.	$[AcF_3]^-$	− 192.22	− 179.70	− 182.30	− 153.68
4.	$[EMIM]^+$	− 116.26	− 77.32	− 113.24	− 79.49
5.	$[BMIM]^+$	− 111.34	− 75.35	− 103.68	− 77.86

Table 7. Experimental solubility of cellulose and hemicellulose in different ILs at different temperatures and time[a,b].

Entry	Ionic liquid type	Condition	Solubility (wt. %)
Microcrystalline cellulose			
1	[EMIM][OAc]	27 °C, 1 h	2.10
		50 °C, 1 h	6.90
		70 °C, 1 h	9.46
		90 °C, 0.5 h	11.38
		90 °C, 1 h	14.63
		90 °C, 2 h	18.93
		110 °C, 1 h	19.10
		130 °C, 1 h	27.69
2.	[BMIM][OAc]	90 °C, 1 h	10.28
		110 °C, 1 h	15.78
3.	[EMIM][$EtSO_4$]	90 °C, 1 h	3.73
		110 °C, 1 h	5.27
4.	[EMIM][$MeSO_3$]	90 °C, 1 h	5.17
		110 °C, 1 h	7.49
5.	[EMIM][SCN]	90 °C, 1 h	2.94
		110 °C, 1 h	4.17
6.	[TMA][$MeSO_4$]	90 °C, 1 h	1.40
		110 °C, 1 h	2.84
Xylan (hemicellulose)			
7.	[EMIM][OAc]	27 °C, 1 h	5.52
		50 °C, 1 h	13.90
		70 °C, 1 h	25.01

[a] The standard uncertainty: u (T) = 0.1 K and u (sol) = 1.21 wt % at 95% confidence level.
[b] Standard uncertainty calculated by using following equation:
$u = (\frac{1}{n(n-1)} \sum_{k=1}^{n} (x_{i,k} - X_i)^2)^{\frac{1}{2}}$.

addition to interaction energy results, COSMO-RS calculations (sigma profiles) also suggest that cellobiose-anti-syn has a qualitatively higher number of polar segments than cellobiose-anti–anti conformer (Figure S9a). This is usually obtained from the cut-off distance for hydrogen bonding [29]. A similar trend was also observed for xylose–glucose conformer (Figure S9b).

In addition, present study interaction energies were compared with the available literature data [37]. The interaction energy between 1,4-O-dimethyl cellobiose and $[OAc]^-$ anion was −178.82 kJ mol^{-1}, which was lower than the cellobiose-(anti–anti)-$[OAc]^-$ interaction energy (−218.23 kJ mol^{-1}). In a similar manner, the results of Guo *et al.* [37] were also compared with cellobiose-anti-syn conformer. The dissimilarity in the interaction energy was mainly due to the number of hydroxyl groups in cellulose moiety. While the 1,4-O-dimethyl cellobiose has seven hydroxyl groups, the cellobiose-anti-syn and cellobiose-anti–anti had eight hydroxyl groups. The interaction energy between the cellobiose-(anti-syn)-anions was much higher than the 1,4-O-dimethyl cellobiose-anions (Table S18). The interaction energy decreasing trend in both the scenarios seems to follow: $[OAc]^- > [Me_2PO_4]^- > [BF_4]^- > [PF_6]^-$. This eventually confirms our QC approach. For theoretical calculations, the consideration of any conformers of cellulose and hemicellulose molecules provides the similar outcome but has variation in interaction energy values.

4.4. Experimental solubility

The experimental solubility of cellulose and hemicellulose was carried out in six different ILs namely [EMIM][OAc], [BMIM][OAc], [EMIM][$MeSO_3$], [EMIM][$EtSO_4$], [EMIM][SCN] and [TMA][$MeSO_4$] at 27–130 °C for 1 h of reaction time. Even though it is a slow process, it still serves our aim to alter the structure of cellulose so as to make it more accessible for enzymatic hydrolysis. A similar dissolution time was undertaken by Balensiefer *et al.* [17] and Vitz *et al.* [27]. The solubility of cellulose and hemicellulose in ILs is reported in Table 7. The solubility of cellulose and hemicellulose in [EMIM][OAc] was found to increase with both time and temperature. [EMIM][OAc] gave a higher solubility of cellulose when compared to [BMIM][OAc]. This follows the results obtained from the interaction energies of cellulose with [EMIM][OAc] and [BMIM][OAc] which were −462.57 and −453.76 kJ mol^{-1}, respectively (Figure 8). Whereas in the case of hemicellulose, the interaction energies for [EMIM][OAc] and [BMIM][OAc] were −509.17 and −503.65 kJ mol^{-1}, respectively. On the other hand, the solvents with higher hydrogen bond basicity (β) value results in a high dissolution

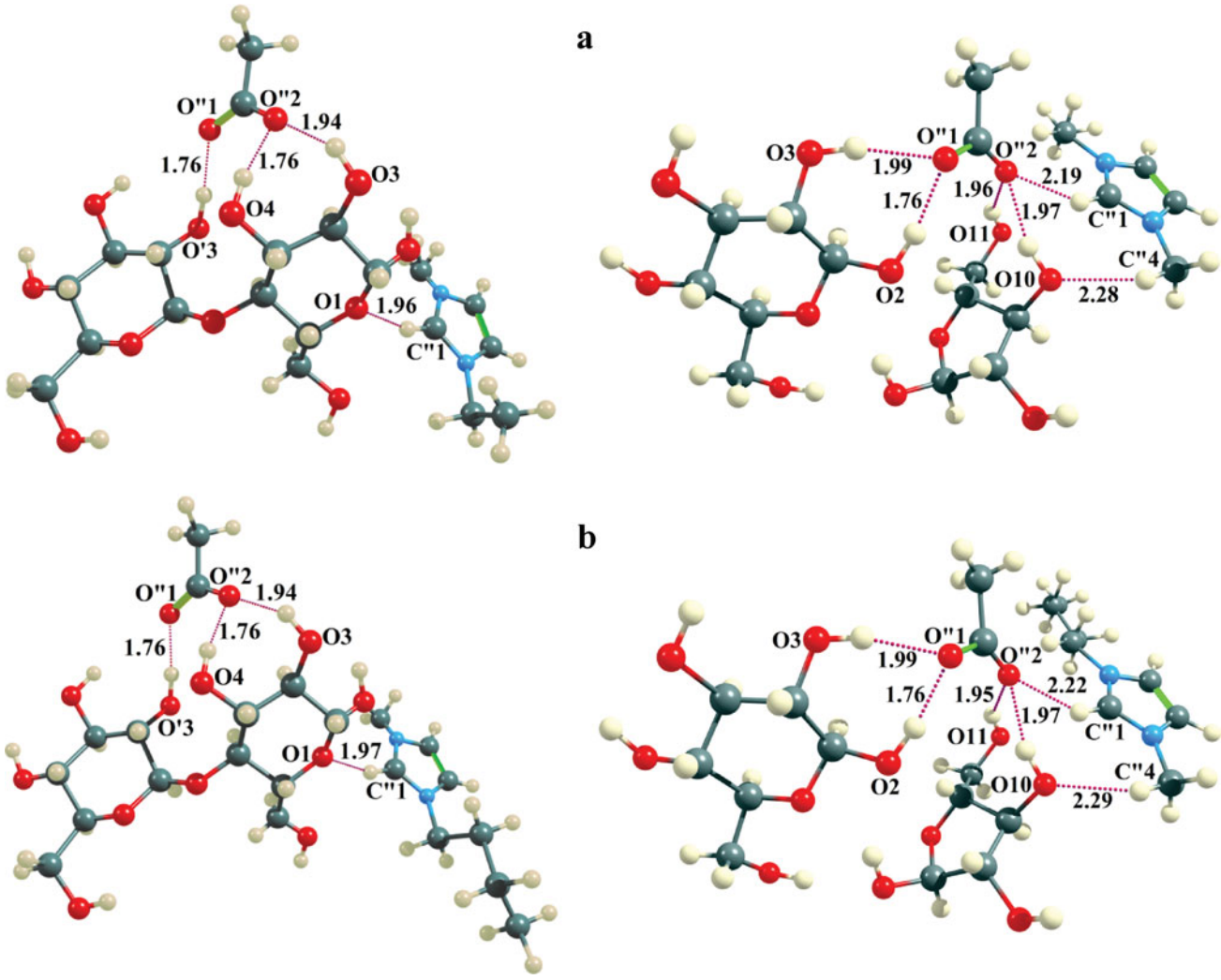

Figure 8. Optimised geometries for cellulose and hemicellulose with IL (a) [EMIM][OAc] and (b) [BMIM][OAc]. The H-bonds are indicated by dotted lines, the bond lengths are in Angstrom (Å), given with corresponding atom numbers (Figure 4).

capacity for a solute [72]. The hydrogen bond basicity of [EMIM][OAc] (1.107) is also more compared to that of [BMIM][OAc] (1.04) [1,40,73]. The characterisations of regenerated cellulose are provided in the supporting information.

In addition, the computed interaction energies from QC (cellulose-IL) and MD (cellulose–anions) simulations are further correlated with the experimental solubility of cellulose (Figure 9). From the figure, it can be observed that the higher the interaction energy between IL and cellulose, the higher the solubility of cellulose. The IL [EMIM][OAc] gave the highest dissolution capacity of cellulose than other ILs due to its stronger interaction energies with cellulose. The dissolution order of anions for the solubility of cellulose in a common cation (i.e. $[EMIM]^+$) was found to be: $[OAc]^- > [MeSO_3]^- > [EtSO_4]^- > [SCN]^-$. The $[OAc]^-$ anion forms strong H-bonds with cellulose which arises due to the presence of higher fraction of polar segments (Figure S10). The $[SCN]^-$ anion exhibits less polarity than oxygen-based anions (Figure S10) thereby resulting in lower interaction energies with cellulose (Table 2). On the other hand, [EMIM][SCN] gave higher dissolution capacity than $[TMA][MeSO_4]$, even though $[MeSO_4]^-$ anion had stronger interactions with cellulose (Table 2). On the contrary, when $[MeSO_4]^-$ combines with $[TMA]^+$ cation, the interaction between cellulose/$[MeSO_4]^-$ gets weakened due to the formation of a strong interaction between $[TMA]^+/[MeSO_4]^-$ (−413.38 kJ/mol; Table 8). The difference between QC and MD simulated interaction energies was found to be significantly higher. This is due to the reason that the QC-based interaction energies were calculated for favourable sites due to the presence of a single ion. The MD-simulated interaction energies included the total molecular system where the interaction of all the ions is accounted. Overall, the interaction energies of

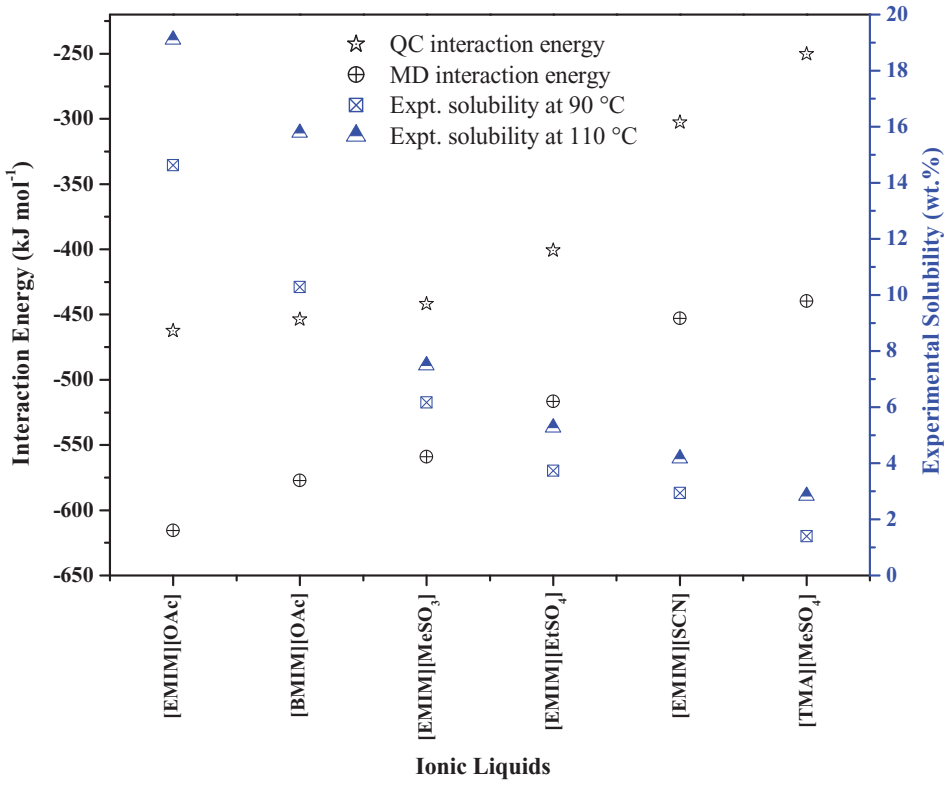

Figure 9. Experimental solubility of cellulose (90 and 110 °C) is correlated with MD and QC interaction energies in different ILs.

IL-cellulose followed a similar pattern in both QC and MD simulations.

Furthermore, the current study was also compared with reported experimental solubility of cellulose in different ILs at different conditions (Table 8). As can be seen from the table, it was observed that there is no major influence of cation on the dissolution of cellulose. As the alkyl chain length in the cation is increased, the solvent power of ILs for cellulose tends to decrease, i.e. [EMIM][OAc] > [BMIM][OAc] > [OMIM][OAc] (Table 8, Avicel, 110 °C). The lower solubility of cellulose in longer chain ILs are primarily due to the reduced strength of acetate anion. This is due to the reduction of hydrophobic interactions of longer chain

Table 8. Experimental solubility of different types of cellulose in several ILs at different conditions.

Ionic liquid	Cellulose type	Condition	Solubility (wt. %)	Reference
[EMIM][OAc]	Avicel	110 °C	15	[74]
	Avicel	100 °C, 35 min	10	[17]
	Cellulose	85 °C	13.5	[22]
	Avicel	100 °C	8	[27]
	Avicel (~DP = 225)	110 °C	28	[74]
	D-glucose	60 °C	60	[74]
	MCC (~DP = 400)	90 °C, 1 h	14.6	Present work
	MCC (~DP = 400)	110 °C, 1 h	19.1	Present work
[EMIM][DEP]	MCC (~DP = 950)	100 °C, 48 h	19.8	[31]
	MCC (~DP = 950)	120 °C, 48 h	19.9	[31]
[EMIM][Cl]	Avicel	90 °C	5	[21]
	Avicel (286)	90 °C	12	[16,19]
[EMIM][Tos]	Avicel	100 °C, 1 h	1	[27]
[EMIM][Br]	Avicel	100 °C, 1 h	1 to 2	[27]
[EMIM][F]	Avicel	100 °C, 1 h	2	[27]
[BMIM][OAc]	Avicel	100 °C, 1 h	12	[27]
[BMIM][SCN]	Pulp cellulose	Microwave	5–7	[14]
[BMIM][Cl]	Avicel	90 °C	5	[21]
[BMIM][HCOO]	Avicel (225)	110 °C	8	[74]
[OMIM][OAc]	Avicel	110 °C	9	[74]

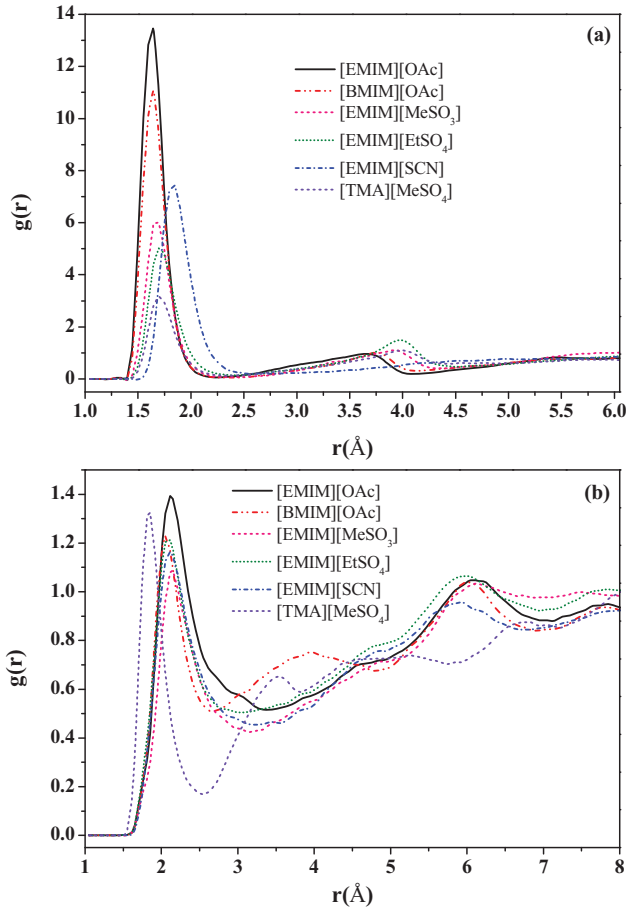

Figure 10. (Colour online) Radial distribution function (RDF) of (a) HS of cellulose and OA of the anion (NA for $[SCN]^-$), (b) OS of cellulose and HC of cation (see Figure S11 for atom notations) the cellulose/IL system.

cations which lowers the ability to screen anion/cellulose complexes [20].

4.5. Molecular dynamics simulations

Apart from the QC calculations and experimental confirmations, the MD simulations were also carried to understand the structural properties of cellulose with ILs. Figure 10(a) shows the RDF between the HS atom of cellulose and OA atom of anions (NA for $[SCN]^-$; see Figure S11 for atom notations). The first solvation shell of RDF plot was located at around 1.65–1.75 Å which indicates a formation of strong H-bond between the cellulose and anions. It should be noted that the corresponding peak for $[SCN]^-$ was obtained at 1.85 Å. In the case of cellulose–cation RDF plot (Figure 10(b)), the peak was obtained at around 2.15 and 1.85 Å for $[EMIM]^+$ and $[TMA]^+$, respectively. It is interesting to note that the ring hydrogen of the cation approaches the cellulose moiety within 2 Å indicating a likely formation of a weak cation–cellulose H-bond. This is further confirmed by the coordination number and interaction energies as given in Table 9, where cellulose is surrounded by six to eight $[OAc]^-$ anions and are present in the first solvation shell (2.35 Å; Figure S12). On the contrary, $[MeSO_3]^-$, $[EtSO_4]^-$, $[SCN]^-$ and $[MeSO_4]^-$ anions had a lower coordination number of 4.91, 4.37, 6.96 and 2.85, respectively. In the case of cellulose-cation interaction, the first solvation shell extends up to 3.35 Å with a coordination number ranging from two to five (Figure S13). Thus, within the first solvation shell, three to eight anions and two to five cations are bounded to cellulose. This fact is supported with the NMR results of Zhang *et al.* [75], where they suggested that the seven ion pairs of ILs are present in the first solvation shell of cellobiose. In addition, it was worthwhile to note that acetate anion formed multiple hydrogen bonds with cellulose molecule (Figures S12 and S14). Similar observations were also reported by Rabideau *et al.* during their study on the influence of water content on the dissolution of cellulose in acetate-based ILs [32]. It was reported that each anion can form up to three to four H-bonds with cellulose.

Table 9 also reports the MD-simulated interaction energies for cellulose–anion and cellulose–cation pairs. From Table 9, it is interesting to note that the strength of the non-bonded interaction energies between anion and cellulose is directly proportional to the experimental solubility (Figure 9). This is a similar trend as observed with QC energies. The simulated energies were in good agreement with the experimental solubility (Figure 9). Apart from cellulose-IL interactions, the cation may diminish the solubility of cellulose by forming H-bonds with anion in the solvation system (Figure S14). Moreover, the obtained MD-simulated interaction energies of cellulose were further compared with the literature data [55,65,76] and shown in Table 10. The table reports the interaction energies for different degree of polymerisations of cellulose molecule with [EMIM][OAc]. From the observation of Table 10, the interaction energies were found to be higher for longer cellulose chains due to the presence of large number of hydroxyl groups. The anion of IL inherits a higher interaction energy with cellulose when compared to cation. The $[OAc]^-$ anion was found to provide a higher interaction energy with cellulose when compared to Cl^- anion [28,64,77]. The reported MD simulations were carried by applying different force field parameters at different production runs. However, the interaction energy results are in line with the variation of the degree of polymerisations which implies that the energy contribution of each monomer is relatively similar. Furthermore, the obtained interaction energy results of ion–cellulose pairs are in agreement with the reported interaction energies (Table 10).

Table 9. Interaction energies (I.E.) and coordination number (C.N.) of the first solvation shell for the different systems obtained from MD simulations[a].

Simulated system	Cellulose–anion		Cellulose–cation		Cation–anion I.E. (kJ mol^{-1})
	I.E. (kJ mol^{-1})	C. N.	I.E. (kJ mol^{-1})	C. N.	
Cellulose-[EMIM][OAc]	− 412.76	7.82	− 202.68	5.18	− 331.64
Cellulose-[BMIM][OAc]	− 401.77	6.47	− 175.33	3.16	− 376.61
Cellulose-[EMIM][$MeSO_3$]	− 370.84	4.91	− 188.20	3.96	− 365.53
Cellulose-[EMIM][$EtSO_4$]	− 333.03	4.37	− 183.61	3.81	− 360.28
Cellulose-[EMIM][SCN]	− 254.66	6.96	− 198.39	4.27	− 303.70
Cellulose-[TMA][$MeSO_4$]	− 230.15	2.85	− 209.43	2.50	− 413.38

[a] Cellobiose-anti-syn as cellulose model.

In summary, the proposed reaction mechanism for the dissolution of cellulose and hemicellulose in imidazolium-based IL ([EMIM][OAc]) is presented in Figure 11. Cellulose molecule has strong inter- and intra-molecular hydrogen linkages through its hydroxyl groups and forms a crystal structure. Upon dissolution of cellulose in [EMIM][OAc], the ionic pairs of IL are partially dissociated into cation and anion, which in turn interacts with the hydroxyl groups of cellulose chain and forms new H-bonds between solute and [EMIM][OAc]. Therefore, the high crystalline order of cellulose gets weakened by disturbing their strong inter- and intra-molecular hydrogen linkages and results in its amorphous structure. From the aforementioned discussions, it is confirmed that the anion of the IL plays a primary role in the dissolution mechanism. As the evidence from RDF plots and coordination numbers, the cation is seen to approach cellulose by forming weak H-bonds.

As a consequence, the dissolution process is seen to follow two opposite pathways. While on one side the cation effects the cellulose solubility due to its strong binding with anion, on the other side anion helps in providing H-bonds with incoming hydroxyl groups of cellulose. A similar mechanism is also applicable for hemicellulose dissolution in IL. However, the proposed reaction mechanism has certain limitations and it is applicable for ILs with highly polar (or strong interacting) anions, i.e. $[OAc]^-$, $[Me_2PO_4]^-$ and $[Cl]^-$ along with slightly polar cations. Therefore, the IL could form new strong H-bonds with cellulose by disturbing their inter- and intra-molecular bonds. If the IL contains both

Table 10. Interaction energies between cellulose and [EMIM][OAc] obtained from MD simulations and compared with literature data[a].

Cellulose type	Cellulose			Production run (ns)	Reference
	[EMIM] $^+$	[OAc] $^-$	[EMIM][OAc]		
Cellulose (DP = 10)[b]	− 975.82	− 2190.15	− 3165.97	4.5	[65]
Glucose[b]	− 97.58	− 219.05	− 316.60	4.5	[65]
Cellulose (DP = 15)[c]	− 1255.32	− 2485.60	− 3740.92	5	[55]
Cellulose (DP = 10)[d]	− 564.30	− 702.24	− 1266.54	250	[76]
Cellulose (DP = 2)[c]	− 202.68	− 412.76	− 615.44	60	Present study
Cellulose (DP = 6; polar method)[e]	− 379.13[f]	− 911.24[g]	− 1290.37[h]	10	[77]
Cellulose (DP = 6; non-polar method)[e]	− 377.87[f]	− 1100.60[g]	− 1478.47[h]	10	[77]
Cellulose (DP = 8)[i]	− 855.26	− 1582.50	− 2437.76	500 [j]	[28]
Cellulose (DP = 8)[i]	− 867.73	− 1373.67[g]	− 2241.40[k]	500 [j]	[28]
Cellulose (DP = 6)[a]	− 500.00	− 1111.12	− 1611.12	100	[64]

[a] Interaction energies are reported per mole of cellulose.
[b] GLYCAM force field parameters are used.
[c] CHARMM force field parameters are used and carried at 300 K.
[d] GLYCAM06 force field parameters are used and carried at 400 K.
[e] GAFF force field parameters are used and carried at 400 K.
[f] 1,3-dimethylimidazolium [C_1mim] cation.
[g] Chloride anion.
[h] [C_1mim]Cl ionic liquid.
[i] GLYCAM force field parameters are used and carried at 373 K.
[J] For single cellulose chain, production run was 10 ns.
[k] [EMIM]Cl ionic liquid.

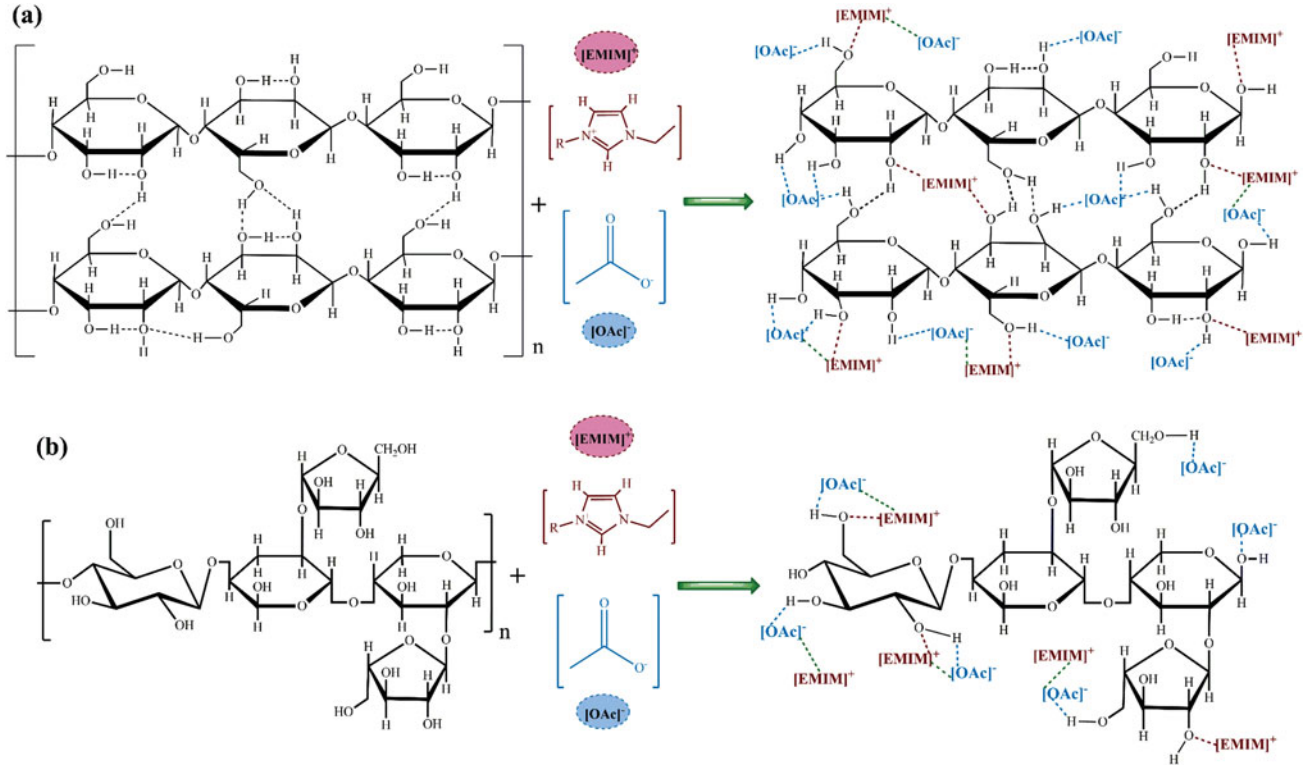

Figure 11. Proposed reaction mechanism for (a) cellulose and (b) hemicellulose dissolution in [EMIM][OAc]. $[OAc]^-$ is seen here to form more H-bonds than $[EMIM]^+$ cation.

polar ions (example: $[TMA]^+$ and $[MeSO_4]^-$), then the dissociation of ionic pairs of ILs would be a challenging task due to the presence strong interactions among the ions which results in lower interactions with cellulose. On the other hand, the IL containing both less polar ions ($[EMIM]^+$ and $[SCN]^-$), the interaction between the cellulose and ions was obviously lower (Table 8). Therefore, the proposed mechanism suggests that combination of highly polar and slightly non-polar ions results in a favourable dissolution and also helps in altering the cellulose structure.

5. Conclusions

The current study reports the basic understanding of the dissolution behaviour of lignocellulosic biomass starting from *ab initio* theory to classical MD. Further to confirm the computational trend, extensive experiments were carried out to validate the dissolution behaviour. Initially, 1428 ILs were screened using COSMO-RS model which is time efficient and computationally inexpensive, thereby allowing us to screen a large number of solvents. After the successful screening of ILs, the potential cation and anion combinations were further screened by QC calculations *via* interaction energy and NBO analysis. Acetate anion and $[EMIM]^+$ cation were found to be good candidates for the dissolution of cellulose and hemicellulose. Furthermore, MD simulations were also performed to investigate the structural properties of the cellulose-IL system. From MD simulations, it was found that the cellulose was bounded with six to seven anions and two to four cations in the first solvation shell. It was also observed that the O-H …O (anion) bond is far stronger than the C-H …O (cation) bonds within cellulose and hemicellulose. From QC and MD simulations, it was noted that acetate anion formed multiple hydrogen bonds with cellulose thereby weaken the crystalline structure of cellulose. Moreover, the effects of anions and cations were studied on the different conformational structures of cellulose and hemicellulose. For theoretical calculations, the consideration of any conformers of cellulose and hemicellulose molecules gives the similar result. The theoretical results were further confirmed by the experimental solubility of cellulose/hemicellulose in ILs which depicted excellent agreement with simulated results. Furthermore, the crystallinity of regenerated cellulose (54.55%) was found to decrease as compared to original cellulose. It was inferred that the anion of the IL is the sole factor in deciding the dissolution mechanism. Incidentally, the cation significantly affects the solubility of cellulose with their acidic protons. This competes on the other side by forming hydrogen bonds with anion. To overcome this issue,

adding a suitable co-solvent to the IL can make the dissolution process economically feasible.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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